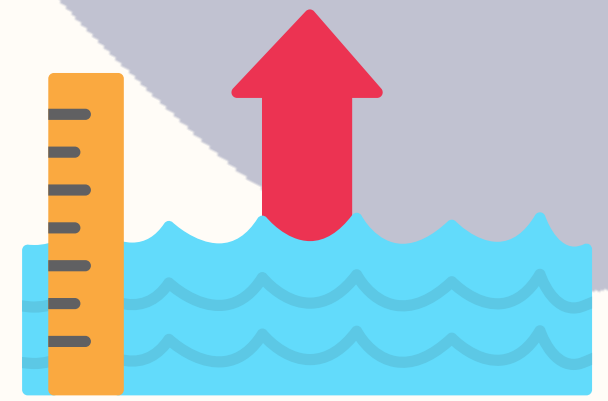


Climate Change Trends Analysis

Are we doomed ?

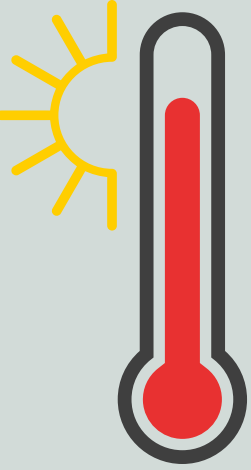
Introduction



Use time-series analysis on temperature, CO2 concentration, and sea-level datasets to predict future climate change trends

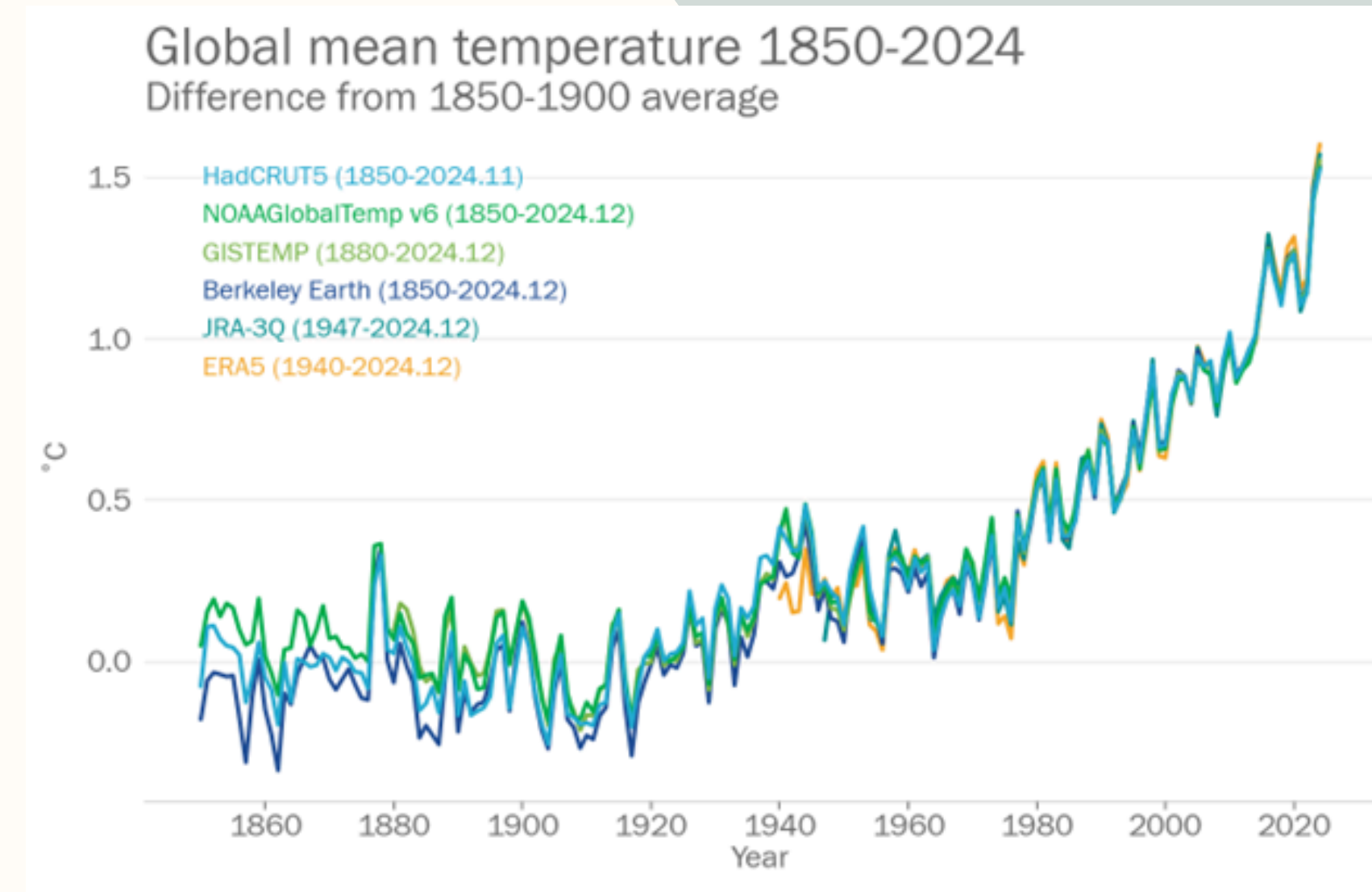


A “hot” topic



Since 1880, annual global temperature has risen at an average rate of 0.08°C per decade

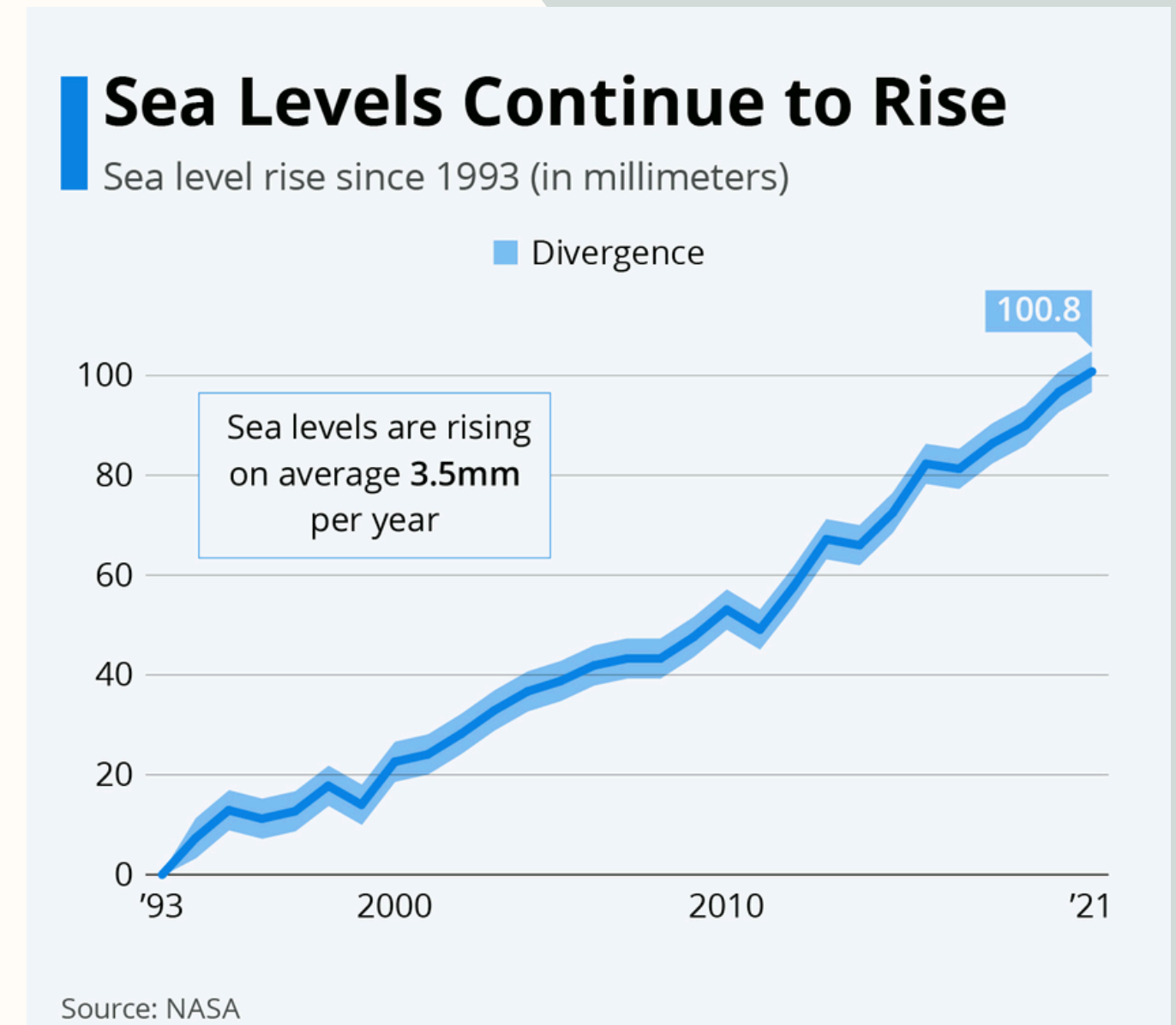
Rate of warming since 1982 is more than two times as fast:
 0.20°C per decade



A “hot” topic

Global mean sea level has risen about 21–24 centimeters since 1880.

The rising water level is mostly due to a combination of melt water from glaciers and ice sheets and thermal expansion of seawater as it warms



Objectives



Visualize the evolution
of climate changes



Predict the future of
climate changes



Realize the situation
and raise awareness

Methodology

Dataset presentations

Daily Temperature of
Major Cities

Daily atmosphere carbon
dioxide concentration

Global Sea Level
1993 - 2021

Methodology



Data processing & Analysis of the variables

Report

Overview

Brought to you by YData

Overview Alerts 4 Reproduction

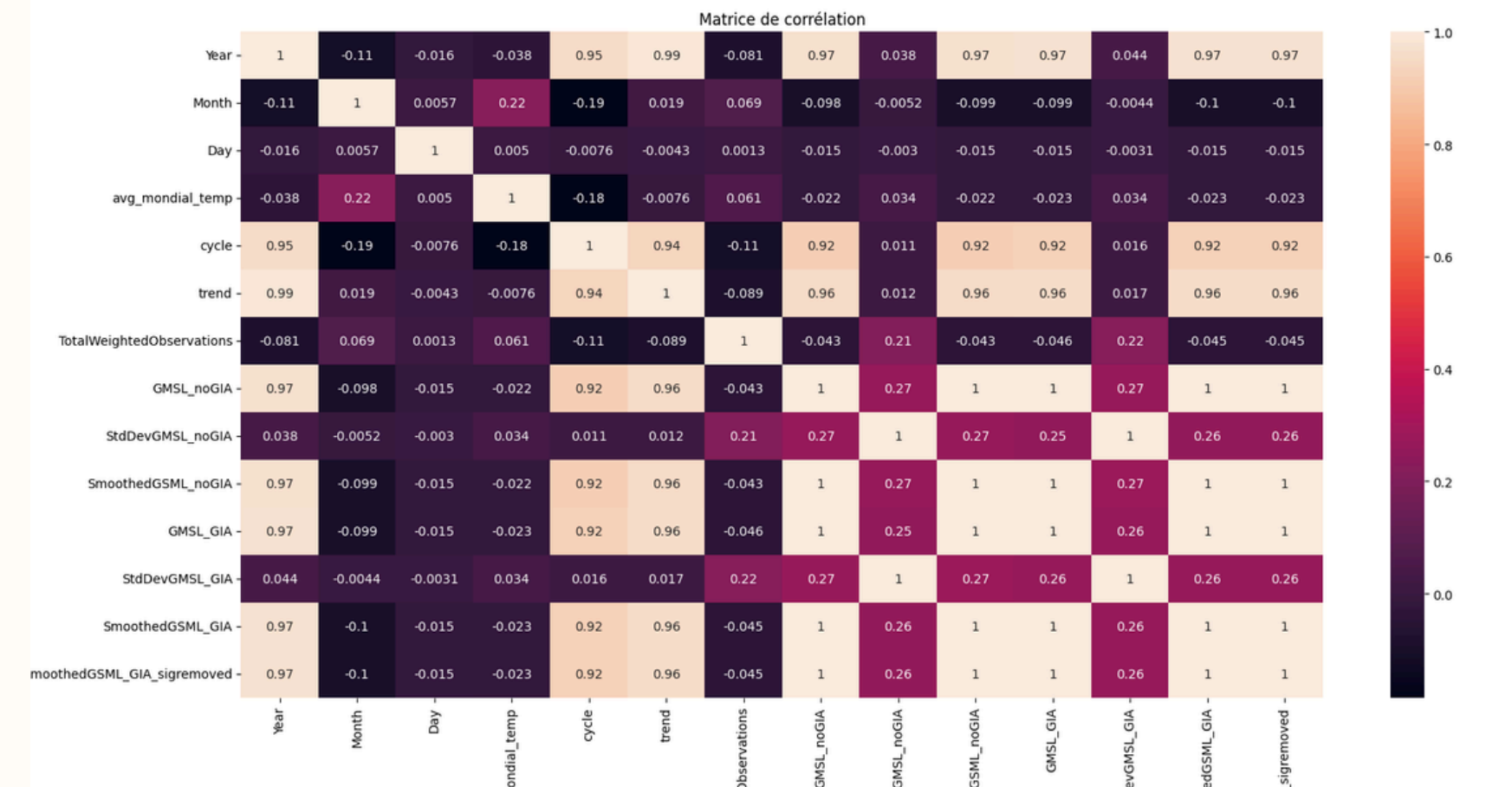
Dataset statistics

Number of variables	7
Number of observations	2672
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	146.3 KiB
Average record size in memory	56.0 B

Variable types

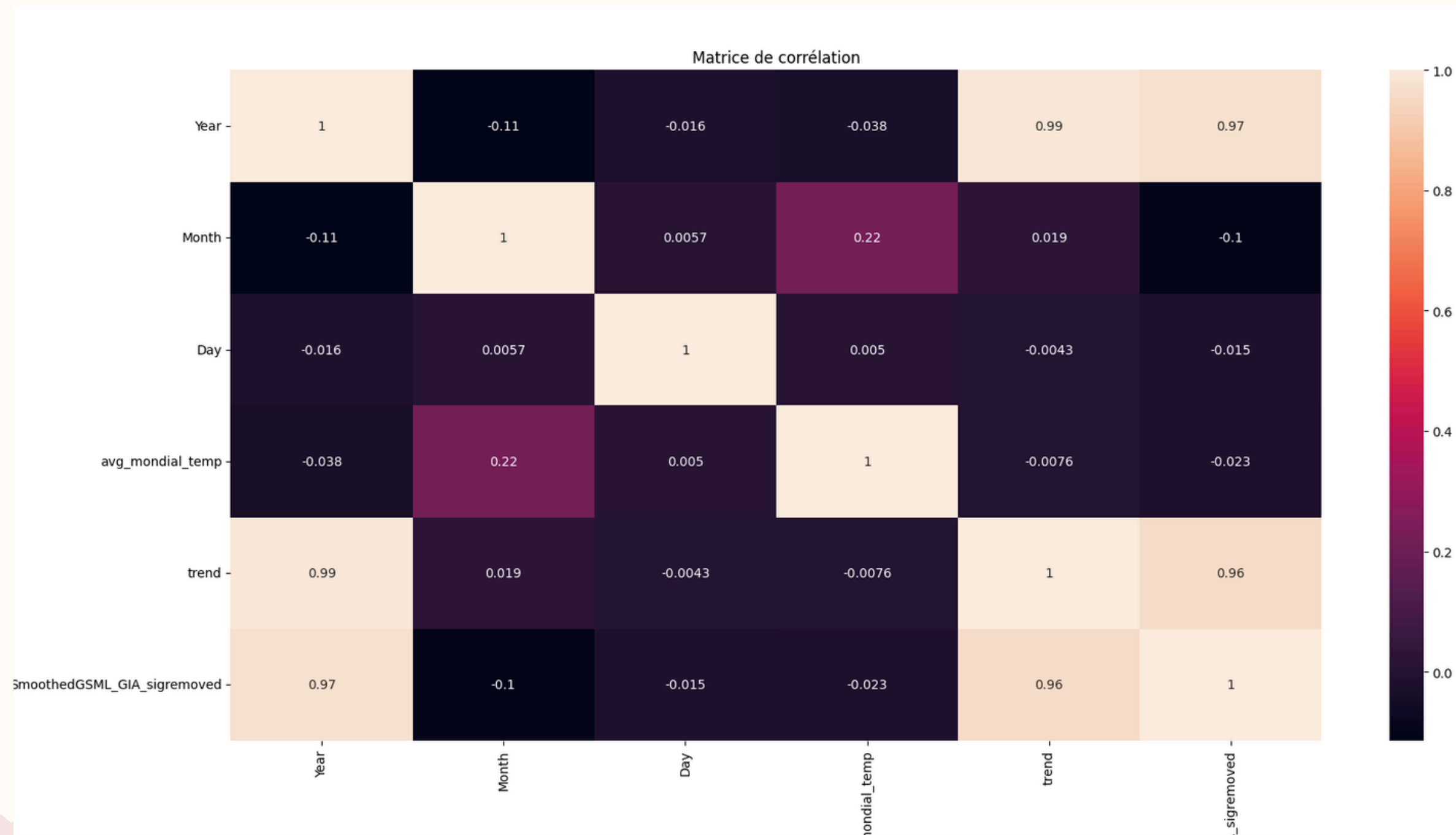
Categorical	1
Numeric	6

Correlation matrix



Methodology

Choosing the variables



Methodology

Testing different models

Linear Regression

The goal of Linear Regression is to predict a continuous value.

The model assumes a linear relationship between the variable that we want to predict and the variables that we use to predict it.

Random Forest

A set of Decision Trees adapt for non-linearity predictions

Less interpretable than linear regression

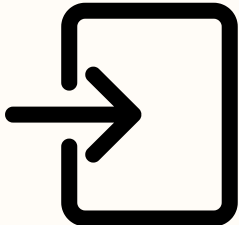
LSTM (Long Short Term Memory)

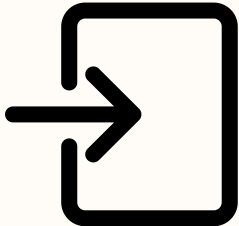
- Variation of RNN
- Appears during the 90's

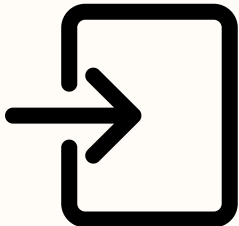
Designed specifically for temporal or sequential data

Can learn patterns over time

3 inputs:

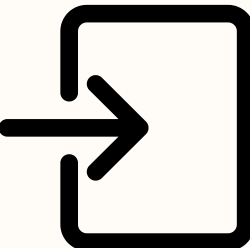
Current input  Provides new information

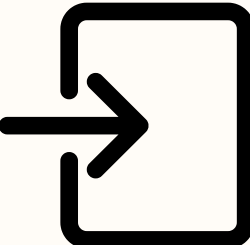
Previous Hidden State  Recent context

Previous Cell State  Persistent information

GRU (Gated Recurrent Units)

Simplified version of LSTM

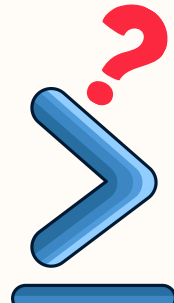
Update gate 

Reset gate 

Comparison with LSTM:

Less complexity
Less parameters
Faster

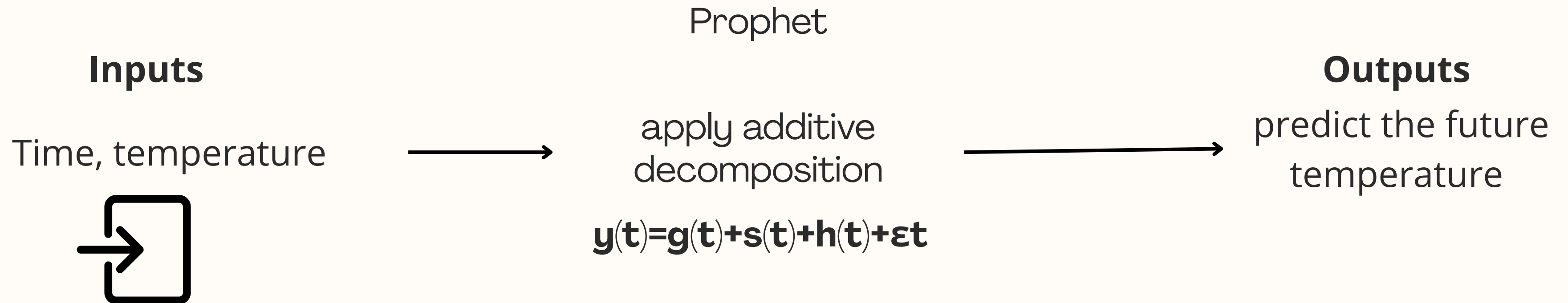
- What do I update with the present?
- What do I keep from my past?

GRU  LSTM

Prophet

- Model developed by Meta
- Time series model,

How it works:



Light Gradient Boosting Machine (LGB)

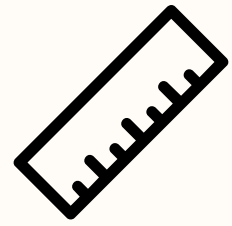
- Supervised machine learning model
- Based on the gradient boosting algorithm

How it works :

- It builds a model tree by tree
- Each new tree tries to correct the mistakes of the previous one thank to a loss function.

Methodology

Challenges



The sizes of the datasets



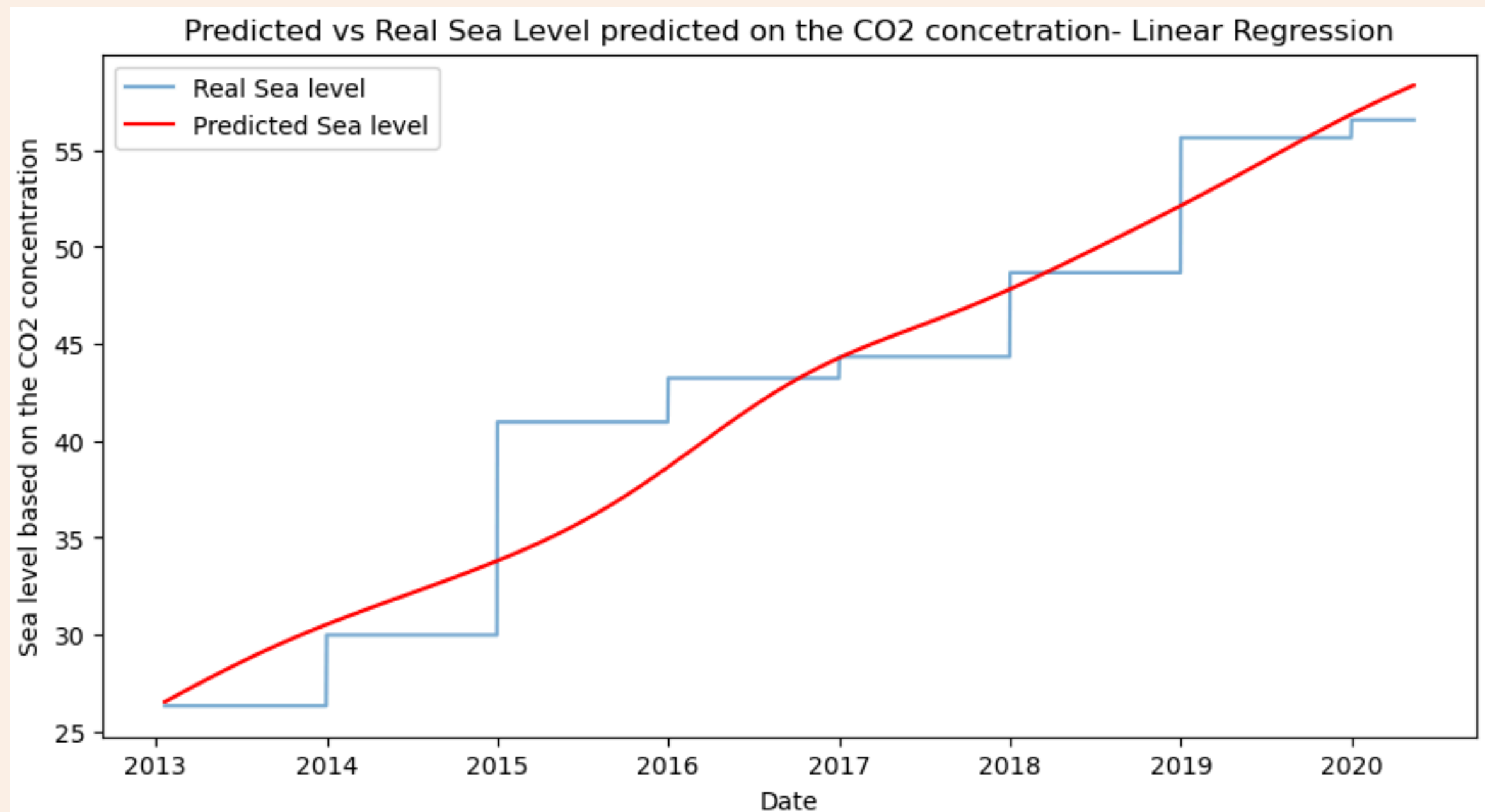
Choosing the models



Time constraint

Findings/Results

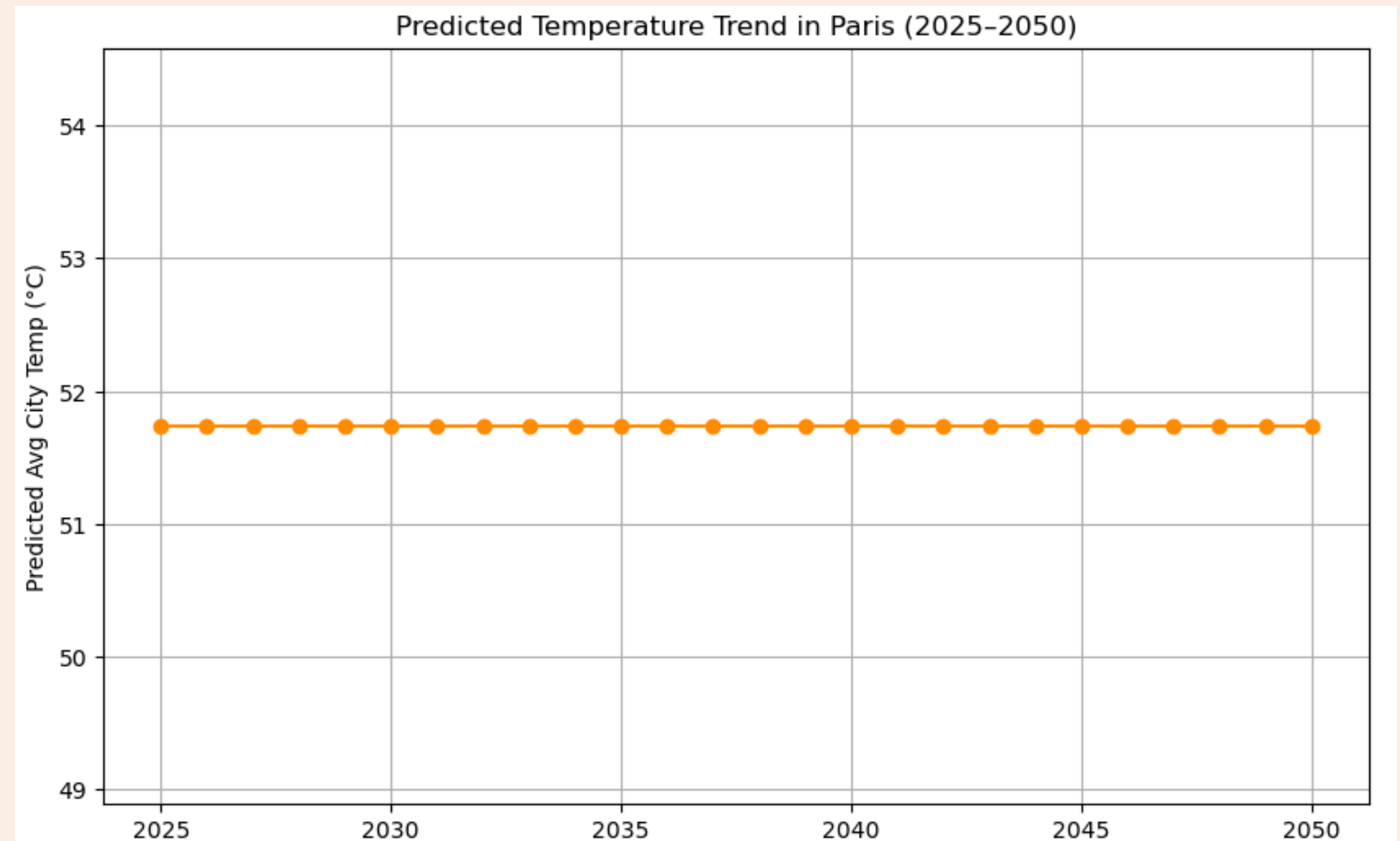
Linear Regression :



Findings/Results

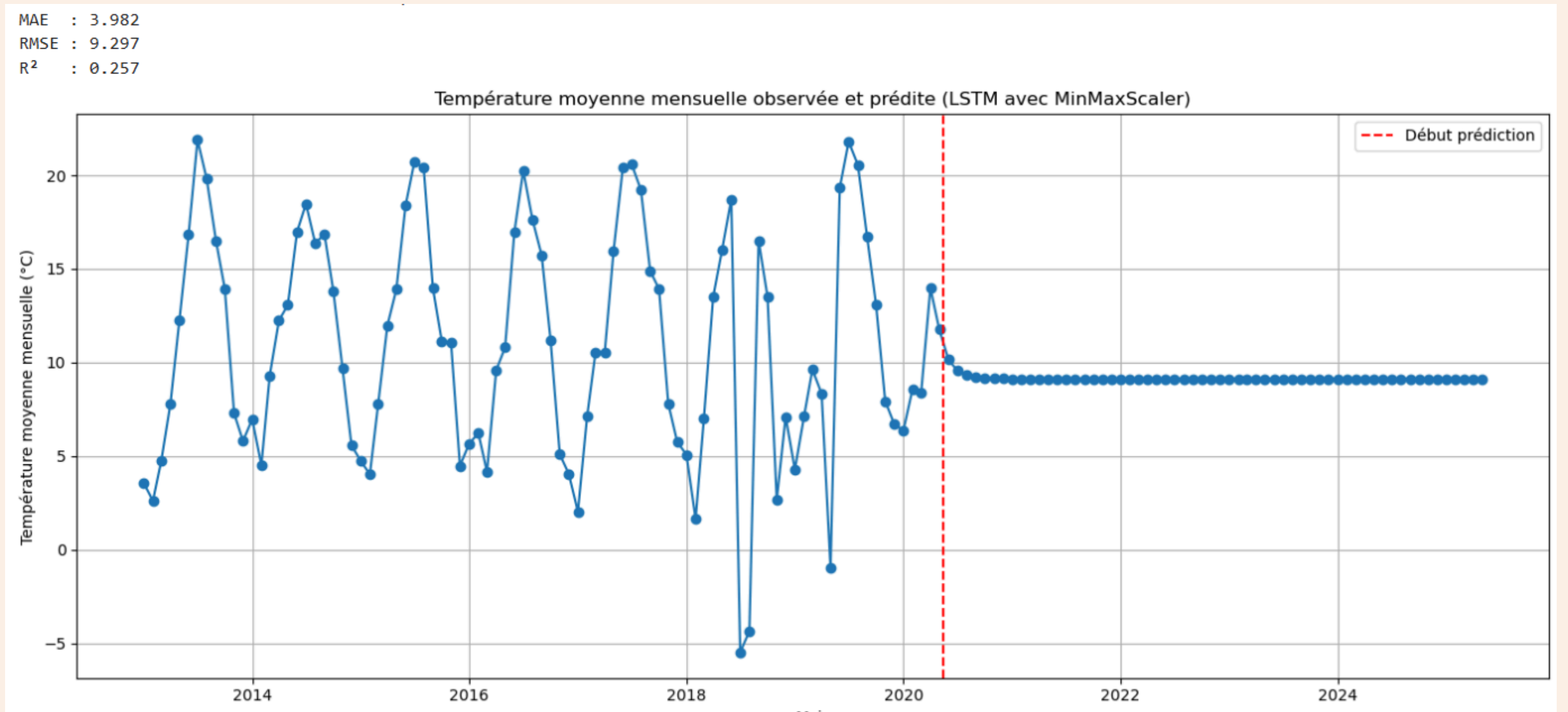
Linear Regression + Random Forest :

	Year	concentration_in_CO2	SmoothedGSML_GIA_sigremoved
0	2025	424.986975	81.305419
1	2026	427.456944	85.761946
2	2027	429.926914	90.218473
3	2028	432.396883	94.675000
4	2029	434.866853	99.131527
5	2030	437.336822	103.588054
6	2031	439.806792	108.044581
7	2032	442.276762	112.501108
8	2033	444.746731	116.957635
9	2034	447.216701	121.414162



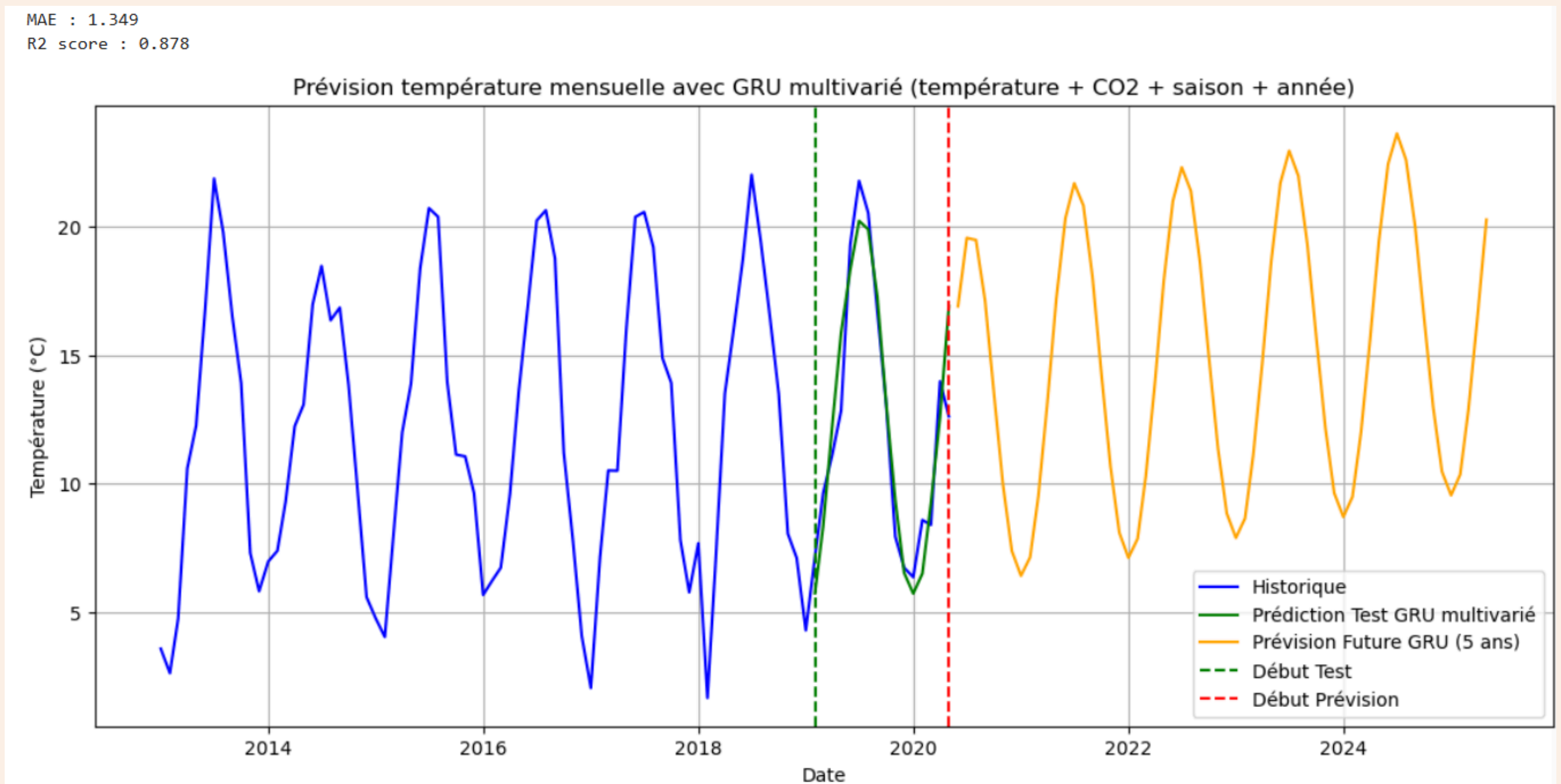
Findings/Results

LSTM:



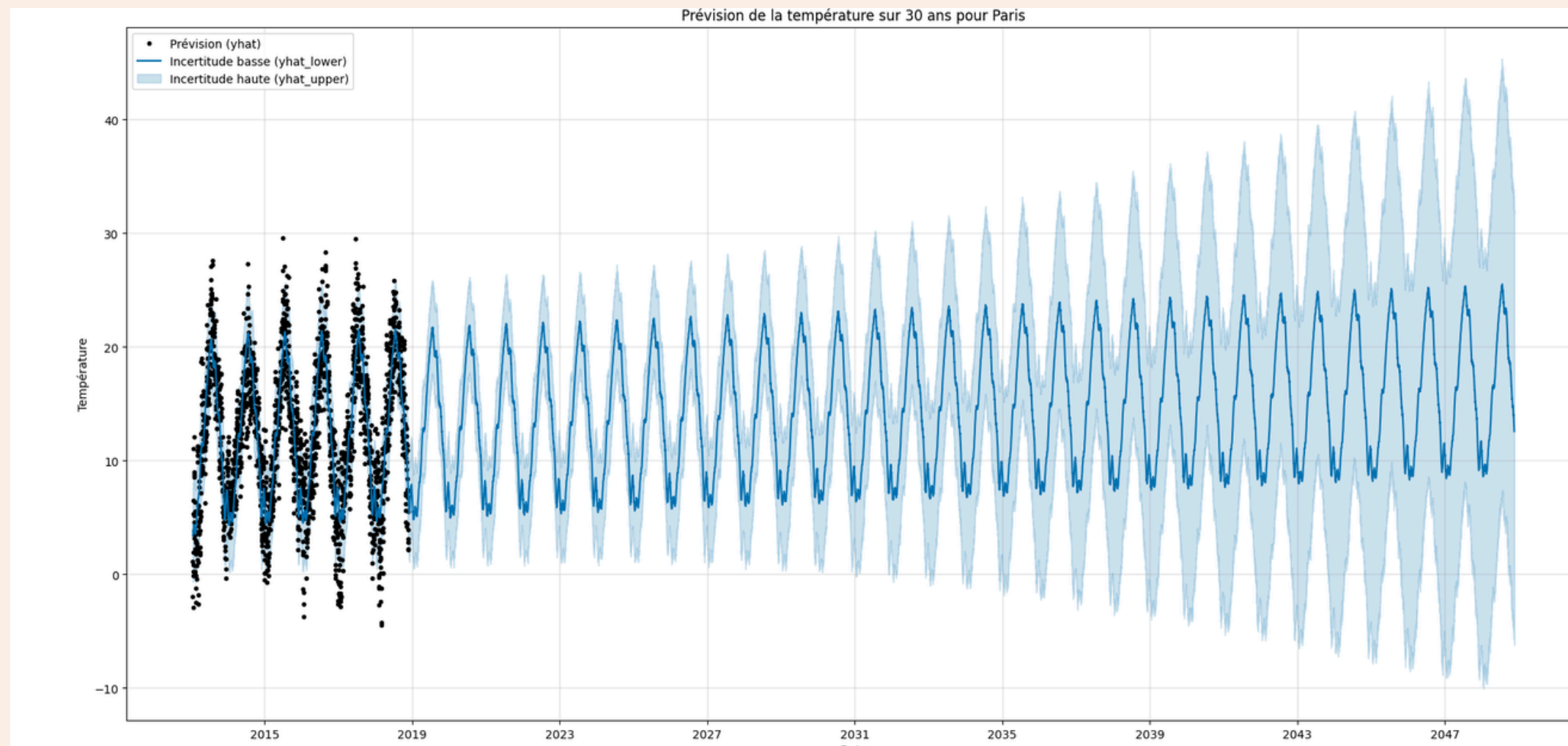
Findings/Results

GRU:



Findings/Results

Prophet :



MAE : 2.69
RMSE : 3.33
 R^2 : 0.70

Findings/Results

LGB :

MAE: 3.0784571796217017

RMSE: 3.3559406461615

Conclusion



Find relevant data on our own

Interpret results on real life data

Work on a significant subject

Helped to be aware of the reality

Conclusion



Find new and more specific models



Optimize data and models



Work with more data



Add new features
(evolution of renewable energies)

Link for GitHub

https://github.com/G1ul1o/Machine_Learning_Project